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Wearable animal information apparatus

Abstract

A sensor apparatus configured for attachment to a collar worn by an animal. The sensor apparatus may include a housing having a first connection feature configured for coupling the sensor apparatus to a first type of animal collar. The first connection feature may be an integral part of the housing. The housing may define a cavity. The sensor apparatus may further include an adaptor assembly configured to be coupled to the housing. The adaptor assembly may include a second connection feature configured for coupling the sensor apparatus to a second type of animal collar that is different from the first type of animal collar. For example, the first type of animal collar may be a nylon collar and the second type of animal collar may be a metal collar. An electronic circuit may be located within the cavity of the housing to store information and/or collect data.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application claims priority to U.S. Provisional Patent Application Ser. No. 63/433,142, filed Dec. 16, 2022, the entirety of which is incorporated herein by reference.

BACKGROUND

(1) Animal collars are used for various reasons, such as to allow an animal to be constrained or controlled by an animal owner. As animal collars are regularly worn by an animal, identification of the animal wearing the animal collar may be provided. Such information may include the name of the animal, pet owner information, as well as other information related to the animal or pet owner. The information may be provided via non-electronic or electronic devices. Typically, however, non-electronic devices are limited in the amount of information that may be conveyed. Further, such devices are subject to degradation over time.

(2) Electronic devices have been provided to overcome some of the deficiencies of the non-electronic tags. For example, electronic devices have been used to store information relating to an animal or an animal owner. Conventional electronic devices, however, are only capable of being used on a specific type of collar, which may reduce their accessibility. Thus, a need exists for a sensor apparatus or wearable animal information apparatus with the capability of being used with different types of animal collars.

BRIEF SUMMARY

(3) A sensor apparatus configured for attachment to a collar worn by an animal. The sensor apparatus may include a housing having a first connection feature configured for coupling the sensor apparatus to a first type of animal collar. The first connection feature may be an integral part of the housing. The housing may define a cavity. The sensor apparatus may further include an adaptor assembly configured to be coupled to the housing. The adaptor assembly may include a second connection feature configured for coupling the sensor apparatus to a second type of animal collar that is different from the first type of animal collar. For example, the first type of animal collar may be a nylon collar and the second type of animal collar may be a metal collar. An electronic circuit may be located within the cavity of the housing to store information and/or collect data.

(4) In one aspect, the invention may be a sensor apparatus configured for attachment to a collar worn by an animal, the sensor apparatus comprising: a sensor apparatus comprising: a housing comprising a first connection feature configured for coupling the sensor apparatus to a first type of animal collar, the housing defining a cavity; and an electronic circuit located within the cavity of the housing, wherein the electronic circuit is configured to at least one of store information and collect data; and an adaptor assembly configured to be coupled to the housing, the adaptor assembly comprising a second connection feature configured for coupling the sensor apparatus to a second type of animal collar that is different from the first type of animal collar;

(5) In another aspect, the invention may be a wearable animal information apparatus comprising: a sensor apparatus comprising: a housing comprising a first connection feature; an adaptor assembly configured to be coupled to the housing, the adaptor assembly comprising a second connection feature; and an electronic circuit located within a cavity of the housing, wherein the electronic circuit is configured to at least one of store information and collect data; a first type of collar configured to be coupled to the sensor apparatus utilizing the first connection feature of the housing, wherein the first type of collar is incapable of attachment to the second connection feature of the adaptor assembly; and a second type of collar configured to be coupled to the sensor

apparatus utilizing the second connection feature of the adaptor assembly, wherein the second type of collar is incapable of attachment to the first connection feature of the housing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present disclosure will become more fully understood from the detailed description and the accompanying drawings, wherein:
- (2) FIG. 1 is a perspective view of a sensor apparatus in accordance with an embodiment of the present invention;
- (3) FIG. 2 is an exploded perspective view of the sensor apparatus of FIG. 1;
- (4) FIG. 3 is an electronic diagram of the electronic components of the sensor apparatus of FIG. 1 in accordance with an embodiment of the present invention;
- (5) FIG. 4A is a top front perspective view of a housing of the sensor apparatus of FIG. 1;
- (6) FIG. 4B is a bottom rear perspective view of the housing of FIG. 4A;
- (7) FIG. 4C is a top plan view of the housing of FIG. 4A;
- (8) FIG. 4D is a cross-sectional view taken along line IV of FIG. 4C;
- (9) FIG. 5A is a perspective view of the housing of FIGS. 4A-4D coupled to a first type of animal collar;
- (10) FIG. 5B is a cross-sectional view taken along line V-V of FIG. 5A;
- (11) FIG. 6A is a front perspective view of a plug of an adaptor assembly of the sensor apparatus of FIG. 1;
- (12) FIG. 6B is a rear perspective view of the plug of FIG. 6A;
- (13) FIG. 7A is a top perspective view of a bracket of the adaptor assembly of the sensor apparatus of FIG. 1;
- (14) FIG. 7B is a top plan view of the bracket of FIG. 7A;
- (15) FIG. 7C is a front view of the bracket of FIG. 7A;
- (16) FIG. 8A is an illustration of the housing and the plugs of the sensor apparatus of FIG. 1 with a dashed line indicating the manner of assembly thereof;
- (17) FIG. 8B is an illustration of the housing with the plugs attached per FIG. 8A, and further illustrating the bracket of the adaptor assembly being coupled thereto;
- (18) FIG. 8C is a bottom perspective view of the sensor apparatus of FIG. 1;
- (19) FIG. 9 is a perspective view of the sensor of FIG. 1 coupled to a second type of animal collar;
- (20) FIG. 10 is a perspective view of a sensor apparatus in accordance with another embodiment of the present invention;
- (21) FIG. 11 is a front view of the sensor apparatus of FIG. 10;
- (22) FIG. 12 is a perspective view of a sensor apparatus in accordance with yet another embodiment of the present invention;
- (23) FIG. 13 is an exploded perspective view of the sensor apparatus of FIG. 12, illustrating a housing and an adaptor assembly thereof;
- (24) FIG. 14 is a close-up view of area XIV of FIG. 13;
- (25) FIG. 15 is a perspective view of the sensor apparatus of FIG. 11 coupled to the second type of animal collar;
- (26) FIG. 16 is a perspective view of a sensor apparatus in accordance with still another embodiment of the present invention;
- (27) FIG. 17 is an exploded perspective view of the sensor apparatus of FIG. 16, illustrating a housing and an adaptor assembly thereof;
- (28) FIG. 18 is a bottom perspective view of the sensor apparatus of FIG. 16;
- (29) FIG. 19 is a perspective view of the sensor apparatus of FIG. 16 coupled to the second type of

animal collar;

(30) FIGS. **20** and **21** illustrate a housing of a sensor apparatus and a band which may be coupled thereto to cover a charging port thereof in accordance with one embodiment of the present invention; and

(31) FIGS. **22** and **23** illustrate a housing of a sensor apparatus and a band which may be coupled thereto to cover a charging port thereof in accordance with another embodiment of the present invention.

DETAILED DESCRIPTION

(32) The following description of the preferred embodiment(s) is merely exemplary in nature and is in no way intended to limit the invention or inventions. The description of illustrative embodiments is intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description. In the description of the exemplary embodiments disclosed herein, any reference to direction or orientation is merely intended for convenience of description and is not intended in any way to limit the scope of the present inventions. Relative terms such as “lower,” “upper,” “horizontal,” “vertical,” “above,” “below,” “up,” “down,” “left,” “right,” “top,” “bottom,” “front” and “rear” as well as derivatives thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) should be construed to refer to the orientation as then described or as shown in the drawing under discussion. These relative terms are for convenience of description only and do not require a particular orientation unless explicitly indicated as such. Terms such as “attached,” “affixed,” “connected,” “coupled,” “interconnected,” “secured” and other similar terms refer to a relationship wherein structures are secured or attached to one another either directly or indirectly through intervening structures, as well as both movable or rigid attachments or relationships, unless expressly described otherwise.

(33) The discussion herein describes and illustrates some possible non-limiting combinations of features that may exist alone or in other combinations of features. Furthermore, as used herein, the term “or” is to be interpreted as a logical operator that results in true whenever one or more of its operands are true. Furthermore, as used herein, the phrase “based on” is to be interpreted as meaning “based at least in part on,” and therefore is not limited to an interpretation of “based entirely on.”

(34) As used throughout, ranges are used as shorthand for describing each and every value that is within the range. Any value within the range can be selected as the terminus of the range. In addition, all references cited herein are hereby incorporated by referenced in their entireties. In the event of a conflict in a definition in the present disclosure and that of a cited reference, the present disclosure controls.

(35) Features of the present inventions may be implemented in software, hardware, firmware, or combinations thereof. The computer programs described herein are not limited to any particular embodiment, and may be implemented in an operating system, application program, foreground or background processes, driver, or any combination thereof. The computer programs may be executed on a single computer or server processor or multiple computer or server processors.

(36) Processors described herein may be any central processing unit (CPU), microprocessor, micro-controller, computational, or programmable device or circuit configured for executing computer program instructions (e.g., code). Various processors may be embodied in computer and/or server hardware of any suitable type (e.g., desktop, laptop, notebook, tablets, cellular phones, etc.) and may include all the usual ancillary components necessary to form a functional data processing device including without limitation a bus, software and data storage such as volatile and non-volatile memory, input/output devices, graphical user interfaces (GUIs), removable data storage, and wired and/or wireless communication interface devices including Wi-Fi, Bluetooth (e.g., Bluetooth classic, Bluetooth low energy), LAN, etc.

(37) In certain embodiments, the present invention may include computer-implemented processes and apparatuses such as processor-based data processing and communication systems or computer

systems for practicing those processes. The present inventions may also include software or computer program code embodied in a non-transitory computer-readable storage medium, which when loaded into and executed by the data processing and communications systems or computer systems, the computer program code segments configure the processor to create specific logic circuits configured for implementing the processes.

(38) Referring now to FIGS. **1** and **2**, a sensor apparatus **1000** is illustrated in accordance with an embodiment of the present invention. The sensor apparatus **1000** may be designed and configured to facilitate its attachment to different types or styles of animal collar. Briefly, an animal collar is a loop-like structure which is configured to be placed around an animal's neck. Often, the collar is used as an attachment point for a leash or lead so that an animal owner can walk with the animal in outdoor environments without fear of the animal becoming loose. The collar may also include physical tags attached thereto containing information about the animal (name), the owner of the animal (name, telephone number, address), and any other information desired to be conveyed. The sensor apparatus **1000** may be attached to the collar as an alternative to, or in addition to, any non-electronic identifying information located on the collar. As described herein, the sensor apparatus **1000** may be designed in a way which enables it to be attached to different types of collars. For example, a nylon collar may be able to be attached to the sensor apparatus **1000** in a specific manner while a metal collar is incapable of being attached to the sensor apparatus **1000** in the same way. Thus, the sensor apparatus **1000** includes features which allow for its attachment to nylon and metal collars, for example, using different structural features of the sensor apparatus **1000**, as described further herein.

(39) The sensor apparatus **1000** may comprise a housing **100** and an adaptor assembly **200** that is configured to be attached to the housing **100**. The housing **100** may be configured for coupling to a first type of animal collar without the adaptor assembly **200**. That is, if it is desired to use the sensor apparatus **1000** with a first type of animal collar (e.g., one formed from a material such as nylon, polyester, neoprene, rubber, elastomer, and leather), this may be accomplished without coupling the adaptor assembly **200** to the housing **100**. However, when it is desired to use the sensor system **1000** with a second type of animal collar (e.g., one formed from metal), then the adaptor assembly **200** may first be coupled to the housing **100** and the second type of animal collar may be coupled to the sensor apparatus **1000** via the adaptor assembly **200** or a portion thereof.

(40) The housing **100** may include a housing body **101** and a housing cover **102**. The housing body **101** may define a cavity **107**, as best shown in FIG. **4D**. An electronic circuit **300** may be located or contained within the cavity. The electronic circuit **300** is depicted generically in FIG. **2** and will be described in greater detail with reference to FIG. **3**. The housing **100** and the electronic circuit **300** may collectively be referred to herein as a sensor assembly. Thus, the sensor apparatus **1000** may comprise a sensor assembly which comprises the housing **100** and the electronic circuit **300** and the sensor apparatus **1000** may further comprise the adaptor assembly **200**.

(41) The housing cover **102** may be detachably coupled to the housing body **101** to selectively open and close an open bottom end of the cavity **107**. The housing body **101** and the housing cover **102** may include various engagement features to facilitate the attachment of the two parts together as is well known in the art. This may be useful for placing various electronic components inside of the cavity and/or for replacing parts as they require replacement. For example, the electronic circuit **300** may include a power source such as a battery which may need to be replaced. In alternative embodiments, the housing **100** may be a singular, monolithic structure and the cavity may not be accessible to a consumer after purchase. That is, in an alternative embodiment the cavity may be sealed closed at the time of manufacture and the electronic components contained therein may not be replaceable. In such an embodiment, the power source of the electronic circuit **300** may be a rechargeable battery which may be capable of being charged with a charging module that can be plugged into a charging port of the sensor apparatus **1000**.

(42) The adaptor assembly **200** may comprise a first plug **210**, a second plug **220**, and a bracket

230. The first and second plugs **210**, **220** may be coupled to the housing **100** and then the bracket **230** may be coupled to the first and second plugs **210**, **220**. In the exemplified embodiment, the bracket **230** may be coupled to the first and second plugs **210**, **220** with fasteners **201**, such as screws. The first and second plugs **210**, **220** may be used to cover up the holes/nesting regions in the housing **100** to prevent use thereof and/or to prevent an animal from getting their teeth caught in the empty space if the first and second plugs **210**, **220** were not used. The first and second plugs **210**, **220** may also be color coded to function as an indication feature. In an alternative embodiment, the bracket **230** may be coupled directly to the housing **100** using fasteners such as screws. In such an embodiment, the plugs **210**, **220** may be omitted, but they also may still be included to provide a seamless housing **100** which may prevent animal teeth getting caught and provide other benefits. As mentioned above, the housing **100** of the sensor apparatus **1000** may be coupled directly to a first type of collar without attaching the adaptor assembly **200** to the housing **100**. The adaptor assembly **200** may be coupled to the housing **100** to enable the sensor apparatus **1000** to be coupled to a second type of collar that is different than the first type of collar in at least one characteristic. Further details about the housing **100**, the plugs **210**, **220**, the brackets **230**, and the electronic circuit **300** will be provided below.

(43) FIG. **3** is a schematic diagram of the components of the electronic circuit **300** which may be housed within the cavity **107** of the housing **100** of the sensor apparatus **1000**. The electronic device **200** may be used to store animal information and/or collect animal data. For example, the electronic device **200** may store information about the animal, including the animal's name, address, owner names, breed, color, size, and other identifying information. The electronic device **200** may collect animal data, which can be any data about the animal including without limitation data about the animal's movement, motion, geographic location, eating habits, sleeping habits, exercise habits, and the like. The electronic circuit **300** may determine, receive (e.g., actively receive, such as collect), and/or transmit information relating to the animal and/or the animal parent, including identification information, location information, medical information, biometric information, movement information, motion information, etc. The information may be real-time information. The electronic circuit **300** may comprise one or more processors, sensors, transponders, etc., including a combination thereof.

(44) The electronic circuit **300** may be attached to a collar as described herein. The electronic circuit **300** may store information related to the animal wearing the collar to which the sensor apparatus **1000** which comprises the electronic circuit **300** is attached and/or track the movement and/or location of the animal. The electronic circuit **300** may comprise, without limitation, an accelerometer **302**, a gyroscope **304**, a Global Positioning System (GPS) **306**, a Wi-Fi Module **308**, a wireless communication module **310** (e.g., cellular, Bluetooth, ultrawide band), a memory **312**, a power source **314**, and a processor **316**. The aforementioned components may be operably coupled together. In one embodiment, each of the components may be coupled to the processor **316** which can dictate the function and operation of the electronic circuit **300**. In other embodiments the various components may be coupled together in other ways. The electronic circuit **300** may also include an indicator such as a speaker, a vibration element, and/or a light element which may be activated to indicate various information to an animal owner, such as battery power remaining, memory storage, data transmission information, and the like. Such indicators may also be used to aid in locating the animal.

(45) The electronic circuit **300** may be configured for radio communication with a base station that may provide the location and/or proximity of the animal. The Global Positioning System (GPS) may be used for locating the animal at a given time or providing information related to the animal's previous whereabouts and geographic locations. The memory **312** may store information about the animal's whereabouts over time. The electronic device circuit **300** may be configured to locate the animal via one or more other tracking methods, such as via Wi-Fi location tracking, cellular location tracking, and the like.

(46) The electronic circuit **300** may communicate with devices and/or objects other than servers and base stations, as described herein. For example, the electronic circuit **300** may communicate with other items, such as items found in a home. Such items may include, for example, pet beds, pet feeders, litter boxes, water bowls, and the like. The electronic circuit **300** may also, or alternatively, include (e.g., store) information relating to the animal and/or pet owner. The information may relate to location information or may be unrelated to location information. The information may be electronically stored and retrieved after the animal is located.

(47) The electronic circuit **300** may identify biometric data of an animal, such as an animal's heart rate, blood pressure, temperature, hydration status, blood oxygen level, etc., to determine and/or transmit a health condition of the animal. The electronic circuit **300** may determine and/or transmit location information of the animal, for example, to determine when the animal is inside/outside of the home of the animal. The electronic circuit **300** may be used to determine environmental conditions related to an animal, for example, the weather, temperature, pollen, etc. of the environment in which the animal is located. The electronic circuit **300** may have a microphone, speaker, camera, and/or recording capability. In examples, camera may be a video/photographic camera or other type of optical sensing device configured to capture images. Camera may be configured to capture single, static images of the area and/or video images of the area. The electronic circuit **300** may have cellular, Bluetooth, Wi-Fi, or other WAN transmission capabilities, which may provide communication capabilities with an external device, such as a smartphone, tablet, server, or the like.

(48) The electronic circuit **300** may communicate with one or more components that are external to the sensor apparatus **1000** via a Bluetooth connection using the wireless communication module. Of course, other wireless communication protocols other than Bluetooth may be used in other embodiments, including without limitation Zigbee, Z-wave, RFID, NFC, or the like.

(49) The accelerometer **302** and/or gyroscope **304** may measure an activity and/or movement of an animal. The gyroscope **304** may measure the animal's orientation. The accelerometer **302** and the gyroscope **304** may together provide 6-axes of motion detection. In some embodiments only one of the accelerometer **302** and gyroscope **304** may be included in the electronic circuit **300**.

(50) A temperature sensor may be included as part of the electronic circuit **300** to measure the animal's body temperature. A GSR Sensor may be included to measure galvanic skin resistance (GSR). For instance, a GSR sensor may measure the amount of sweat or moisture detected on the body of the animal. The GPS **306** may identify and/or determine the location (e.g., current location or past locations) of the animal. The memory **312** may be of any size. The memory **312** may be coupled to each of the sensors to store data obtained by the sensors. The memory **312** may be operably coupled to the processor **316** and/or to a transmitter to transmit the data obtained by the sensors to an external device. The external device may be a software application residing on a computer, a smart phone, or the like.

(51) The power source **314** may comprise one or more batteries which may be contained within (e.g., self-contained within) the cavity **107** of the housing **100**. The power source **314** may supply power to each of the components of the electronic circuit **300**. The power source **314** may be a rechargeable battery that can be charged by a power cord that is coupled to a wall outlet or a power source of a vehicle or the like. The power source **314** may be removable and/or modular. The power source **314** may be chargeable while the sensor apparatus **1000** is attached to the animal. For example, an interface may be provided through the housing **100** of the electronic circuit **300** that may allow the power source **314** to be charged while sensor apparatus **1000** is attached to a collar that is placed around the animal's neck. The interface may be, for example, a USB interface, induction pins, wireless charging, wireless communications, etc., that allow the power source **314** to be accessed and charged while the sensor apparatus **1000** is attached to a collar. Although a Li-Po battery may be used, such battery is for illustration purposes only and any type of battery may be used.

(52) The electronic circuit **300** may store, receive (e.g., actively receive), and/or transmit identification information of the animal and/or pet owner. For example, the processor **316** may store, receive (e.g., actively receive), and/or transmit real-time information of the animal and/or pet owner. The processor **316** may be configured to translate and/or store data from components housed within electronic circuit **300** (e.g., accelerometer **202**, gyroscope **206**, magnetometer, GPS **306**, etc.). The processor **316** may be configured to permit electronic circuit **300** of the sensor apparatus **1000** to function and/or assist in one or more modes, such as active mode, sleep mode, transmit mode, onboarding mode. The processor **316** may identify the animal and/or the animal owner, track the location of the animal, monitor biometric information of the animal or activity (e.g., heart rate, steps, calories burned, etc.) of the animal, and/or environmental conditions related to an animal. The electronic circuit **300** may identify such information via one or more sensors, such as accelerometers, gyroscopes, temperature sensors, heart rate sensors, magnetometers, electrocardiogram (EKG, otherwise known as ECG) electrodes, photoplethysmography (PPGs) and/or reflection mode PPGs (PPGr) sensors, or one or more other sensors of electronic circuit **300** or external to electronic circuit **300** that detect information of an animal. For example, the electronic circuit **300** may identify biometric data of the animal via biosensors to determine health indicators of the animal, such as glucose, cortisol, serotonin, scrum symmetric dimethylarginine (SDMA), and other indicators of the animal. The processor **316** may track such data over certain time periods. As an example, the processor **316** may be an ARM Cortex M0-M3, or the like.

(53) The electronic circuit **300** may include one or more components, such as a light source. The light source may be an LED light source, for example. The light source may be indicators. The light source may provide one or more indicators having one or more colors and other effects. For example, light source may have one or more light indicators that may be different colors, the same colors, or a combination of the two. The light indicators may blink at speeds (e.g., fast intervals, slow intervals), etc. The light indicators may represent aspects of the animal, such as the animal's activity (e.g., eating/drinking activity, sleeping activity, exercise activity, etc.). The light indicators may represent aspects of the environment surrounding sensor apparatus **1000**, such as the weather, precipitation, moisture, humidity, and the like. The light indicators may represent aspects of sensor apparatus **1000**, such as an indication of sufficient or lack of power of the power source **314**, of the power source **314** being charged, and the like. The light indicators may represent data receipt and/or transmission. For example, the light indicators may represent successful data receipt and/or transmission and/or unsuccessful data receipt and/or transmission. Instead of or in addition to light indicators, the electronic circuit **300** may have speakers or vibration elements to achieve the indication functions described herein.

(54) Thus, the electronic circuit **300** of the sensor apparatus **1000** is configured to at least one of store information and collect data using the various sensors, power sources, processors, and the like. The stored information may be information about the animal wearing the sensor apparatus **1000** or that animal's owner. The collected data may include data about the animal's geographic locations at a specific time or over a period of time, data about the animal's movements, data about the animal's biometrics and daily habits, and the like. There are numerous functions which may be achieved by the electronic circuit **300** and the invention described herein is not intended to be limited by those functions in all embodiments.

(55) With reference to FIGS. 4A-4C, the housing **100** will be described. The housing **100** comprises the housing body **101** and the housing cover **102** as previously described. Furthermore, the housing body **101** comprises a front surface **103**, a rear surface **104**, a top surface **105**, and a bottom surface **106**. The housing body **101** may comprise a main body portion **110** and a first connection feature **150**. The main body portion **110** and the first connection feature **150** may be integrally connected and formed as a singular monolithic component. The main body portion **110** may comprise a first end **111** and a second end **112** each of which extends generally between the front and rear surfaces **103**, **104**. The first connection feature **150** may comprise a first connector

160 extending from the first end **111** of the main body portion **110** and a second connector **180** extending from the second end **112** of the main body portion **110**. The first and second connectors **160**, **180** of the first connection feature **150** may both be utilized to facilitate attachment of the housing **100** to a first type of animal collar, as described herein.

(56) The first connector **160** may comprise a first flange **161** extending from the first end **111** of the main body portion **110** at a position that is adjacent to the front surface **103** of the housing body **101** and a second flange **162** extending from the first end **111** of the main body portion **110** at a position that is adjacent to the rear surface **104** of the housing body **101**. The first flange **161** may have an inner surface **163** and an outer surface **164**, with the outer surface **164** being generally flush with the front surface **103** of the housing body **101**. The second flange **162** may have an inner surface **165** and an outer surface **166**, with the outer surface **166** being generally flush with the rear surface **104** of the housing body **101**. The first and second flanges **161**, **162** may extend generally perpendicularly from the first end **111** of the main body portion **110** in a spaced apart manner with the inner surfaces **163**, **165** facing each other and being spaced apart from each other. The various flanges described herein may have varying widths to accommodate different size collars, from small/thin collars to very wide collars.

(57) A first arm **167** may extend from the inner surface **163** of the first flange **161** in a direction towards the second flange **162**. The first arm **167** may extend from the first flange **161** at a terminal end of the first flange **161** which is located furthest from the first end **111** of the main body portion **110**. The first arm **167** may terminate in a distal end **168**. The first arm **167** may be oriented perpendicularly relative to the first flange **161**. A second arm **169** may extend from the inner surface **165** of the second flange **162** in a direction towards the first flange **161**. The second arm **169** may extend from the second flange **162** at a terminal end of the second flange **162** which is located furthest from the first end **111** of the main body portion **110**. The second arm **169** may terminate in a distal end **170**.

(58) The distal ends **168**, **170** of the first and second arms **167**, **169** may be spaced apart from one another by a gap **174**. The first arm **167** comprises an inner surface **171** that faces the first end **111** of the main body portion **110** of the housing **100** and the second arm **169** comprises an inner surface **172** that faces the first end **111** of the main body portion **110**. The inner surfaces **171**, **172** of the first and second arms **167**, **169** may be spaced apart from the first end **111** of the main body portion **110** such that a first nesting region **175** is defined between the first and second arms **167**, **169** of the first connector **160** and the first end **111** of the main body portion **110** of the housing **100**. A portion of a collar may be positioned within the first nesting region **175** when the housing **100** is coupled to the collar. The inner surfaces **171**, **172** of the first and second arms **167**, **169** may comprise gripping protrusions **173** extending inwardly towards the first end **111** of the main body portion **110**. The gripping protrusions **173** may contact the collar to which the housing **100** is attached to prevent the housing **100** from moving substantially along the collar when in use. The gripping protrusions **173** may be formed at or adjacent to the distal ends **168**, **170** of the first and second arms **167**, **169**, respectively. In an embodiment, the gripping protrusions **173** may include one or more stepped portions, each having an apex configured to contact the collar **10** in an assembled configuration to inhibit lateral movement of the collar **10** relative to the housing **100**. Alternatively, the gripping protrusions **173** may have a smooth linear or arcuate inner geometry and/or may be supplemented with a coating, material or finish configured to enhance grip.

(59) The second connector **180** may comprise a third flange **181** extending from the second end **112** of the main body portion **110** at a position that is adjacent to the front surface **103** of the housing body **101** and a fourth flange **182** extending from the second end **112** of the main body portion **110** at a position that is adjacent to the rear surface **104** of the housing body **101**. The third flange **181** may have an inner surface **183** and an outer surface **184**, with the outer surface **184** being generally flush with the front surface **103** of the housing body **101**. The fourth flange **182** may have an inner surface **185** and an outer surface **186**, with the outer surface **186** being generally flush with the rear

surface **104** of the housing body **101**. The third and fourth flanges **181**, **182** may extend generally perpendicularly from the second end **112** of the main body portion **110** in a spaced apart manner with the inner surfaces **183**, **185** facing each other and being spaced apart from each other.

(60) A third arm **187** may extend from the inner surface **183** of the third flange **181** in a direction towards the fourth flange **182**. The third arm **187** may extend from the third flange **181** at a terminal end of the third flange **181** which is located furthest from the second end **112** of the main body portion **110**. The third arm **187** may terminate in a distal end **188**. The third arm **187** may be oriented perpendicularly relative to the third flange **181**. A fourth arm **189** may extend from the inner surface **185** of the fourth flange **182** in a direction towards the third flange **181**. The fourth arm **189** may extend from the fourth flange **182** at a terminal end of the fourth flange **182** which is located furthest from the second end **112** of the main body portion **110**. The fourth arm **189** may terminate in a distal end **190**.

(61) The distal ends **188**, **190** of the third and fourth arms **187**, **189** may be spaced apart from one another by a gap **194**. The third arm **187** comprises an inner surface **191** that faces the second end **112** of the main body portion **110** of the housing **100** and the fourth arm **189** comprises an inner surface **192** that faces the second end **112** of the main body portion **110**. The inner surfaces **191**, **192** of the third and fourth arms **187**, **189** may be spaced apart from the second end **112** of the main body portion **110** such that a second nesting region **195** is defined between the third and fourth arms **187**, **189** of the second connector **180** and the second end **112** of the main body portion **110** of the housing **100**. A portion of a collar may be positioned within the second nesting region **195** when the housing **100** is coupled to the collar. The inner surfaces **191**, **192** of the third and fourth arms **187**, **189** may comprise gripping protrusions **193** extending inwardly towards the second end **112** of the main body portion **110**. The gripping protrusions **193** may contact the collar to which the housing **100** is attached to prevent the housing **100** from moving substantially along the collar when in use. The gripping protrusions **193** may be formed at or adjacent to the distal ends **188**, **190** of the third and fourth arms **187**, **189**, respectively, and may be formed substantially similar to the gripping protrusions **173**.

(62) The first and second arms **167**, **169** collectively form a first arm structure which is spaced apart from the first end **111** of the main body portion **110** of the housing **100** by the first nesting region **175**. The first nesting region **175** is bounded by the first end **111** of the main body portion **110**, the inner surfaces **171**, **172** of the first and second arms **167**, **169**, and the inner surfaces **163**, **165** of the first and second flanges **161**, **162**. The third and fourth arms **187**, **189** collectively form a second arm structure which is spaced apart from the second end **112** of the main body portion **110** of the housing **100** by the second nesting region **195**. The second nesting region **195** is bounded by the second end **112** of the main body portion **110**, the inner surfaces **191**, **192** of the third and fourth arms **187**, **189**, and the inner surfaces **183**, **185** of the third and fourth flanges **181**, **182**.

(63) With reference to FIG. 4D, as mentioned above the housing body **101** defines a cavity **107** that houses the electronic circuit **300** of the sensor apparatus **1000**. As mentioned previously, the housing cover **102** may be detachable from the housing body **101** to expose the cavity **107**, although this is not required in all embodiments. As best seen in FIG. 4D, the bottom surface **106** of the housing body **101** may be arcuate, and more specifically concave. This may ensure that the bottom surface **106** of the housing body **101** fits naturally around the curvature of the animal's neck when the animal is wearing a collar with the housing **100** attached. The bottom surface **106** of the housing body **101** may be flat/planar in other embodiments, one of which will be described later on in this document.

(64) FIGS. 5A and 5B illustrate the housing **100** of the sensor apparatus **1000** coupled to a first type of animal collar **10**. More specifically, the sensor assembly which comprises the housing **100** and the electronic circuit **300** is coupled to the first type of animal collar **10**. That is, when the first type of animal collar **10** is used, the housing **100** may be coupled thereto without the adaptor assembly **200** being first coupled to the housing **100**. In fact, in some embodiments, when the adaptor

assembly **200** is coupled to the housing **100**, the first type of animal collar **10** may be prevented from being coupled to the housing **100** or the sensor apparatus **1000**. As noted previously, the first type of animal collar **10** may be formed from a material selected from the group consisting of nylon, polyester, neoprene, rubber, elastomer, and leather, including combinations of these materials, or these materials which may be coated or covered or combined with other materials. In some embodiments, the first type of animal collar **10** may be formed from a fabric material.

(65) The first type of animal collar **10** may be coupled to the housing **100** by wrapping over and around the arm structures and extending along the bottom of the housing **100**. More specifically, a first portion **11** of the first type of animal collar **10** may nest within the first nesting region **175** of the housing **100**, a second portion **12** of the first type of animal collar **10** may nest within the second nesting region **195** of the housing **100**, and a third portion **13** of the first type of animal collar **10** that extends between the first and second portions **11**, **12** of the first type of animal collar **10** may extend along the bottom of the housing **100**. Thus, the first type of animal collar **10** may be coupled to the housing **100** via the first connection feature **150** of the housing **100** which comprises the first and second connectors **160**, **180** as described in detail above.

(66) The first type of animal collar **10** may comprise a clasp mechanism **15** to facilitate attaching the first type of animal collar **10** onto or detaching the first type of animal collar **10** from the animal. The clasp mechanism **15** may be a conventional buckle with male and female parts that engage one another. The clasp mechanism **15** may alternatively take on other forms, including being a breakaway mechanism that automatically separates upon application of a force to prevent choking. The exact structure and type of the clasp mechanism **15** is not to be limiting of the invention in all embodiments. When the clasp mechanism **15** is in an attached state as shown in FIG. 5A, the collar **10** forms a loop structure that is configured to be positioned around an animal's neck. Furthermore, the housing **100** is supported on the first type of animal collar **10** to perform its functions as described below.

(67) Thus, the housing **100** may be a one-piece, monolithic structure which can be used by itself (i.e., without the adaptor assembly **200**) when the sensor apparatus **1000** is being coupled to the first type of animal collar **10**. The first connection feature **160** is integral with the housing **100** and therefore the housing **100** includes all features required to attach the sensor apparatus **1000** to the first type of animal collar **10**.

(68) In order to render the sensor apparatus **1000** capable of being coupled to a second type of animal collar as described further herein below, the adaptor assembly **200** (or at a minimum, the bracket member **230** of the adaptor assembly **200**) may need to first be coupled to the housing **100**. FIGS. 6A and 6B are front and rear perspective views of the first and second plugs **210**, **220** of the adaptor assembly **200**. The first and second plugs **210**, **220** are identical, and thus while the description below references only the first plug **210**, it is applicable to both of the first and second plugs **210**, **220**. The first plug **210** may be configured to be positioned within the first nesting region **175** of the housing **100** and the second plug **220** may be configured to be positioned within the second nesting region **195** of the housing **100**, as described in greater detail below.

(69) The first plug **210** comprises a front **211**, a rear **212**, a top **213** and a bottom **214**. The rear **212** may be flat to enable it to rest in abutting contact with the first end wall **111** of the main body portion **110** of the housing **100** when coupled thereto. The front **211** may have various contours necessary to enable the first plug **210** to engage with the first and second arms **171**, **172** of the first connector **160** when the first plug **210** nests within the first nesting region **175**. The front **211** may have a top portion **215** that extends from the top **213** to a transition region **216** and a bottom portion **217** that extends from the transition region **216** to the bottom **214**. The top portion **215** may be planar in some embodiments, although it is arcuate, and more specifically convex in the exemplified embodiment in order to match the contour of the portions of the housing **100** that are adjacent to the first plug **210** when the first plug **210** is positioned within the first nesting region **175**. The bottom portion **217** may be arcuate, and more specifically concave, along at least a

portion thereof. An upper region of the bottom portion **217** may form a downwardly facing shoulder **225** which may engage or contact the arm structure of the housing **100** to retain the first plug **210** in place when assembled/coupled to the housing **100**.

(70) A central region **218** of the bottom portion **217** may comprise a recess **219**. Within the recess **219**, there may be ribs **222** that are configured to mate with the gripping protrusions **173** of the housing **100** described above. Furthermore, there may be a protrusion **221** protruding from the recess **219**. The protrusion **221** may protrude beyond the outer surface of the bottom portion **217** of the front **211**. The protrusion **221** may be located centrally along the recess **219** such that there are two portions of the recess **219** located on opposing sides of the protrusion **221**. Thus, when the first plug **210** is positioned within the first nesting region, the protrusion **221** extends into the gap **174** between the first and second arms **167**, **169** of the first connector **160** and the gripping protrusions **173** on the inner surfaces of the first and second arms **167**, **169** are aligned with the two portions of the recess **219**. The first plug **210** may be specifically sized and shaped to fit snugly within the first nesting region **175**. The second plug **220** may be specifically sized and shaped to fit snugly within the second nesting region **195**.

(71) The bottom **214** of the first plug **210** may comprise two apertures **223**. While two apertures **223** are shown in the exemplified embodiment, there may be one aperture or more than two apertures in other embodiments. The apertures **223** may be configured to receive the screws **201** mentioned above with reference to FIG. **2** to facilitate coupling of the bracket **230** thereto.

(72) Referring to FIGS. **7A-7C**, the bracket **230** will be described. The bracket **230** comprises a plate member **231**. The plate member **231** may have a rectangular shape as shown, although it may take on other shapes in other embodiments, including square, triangular, or other polygonal shapes. The plate member **231** may define or form a frame which surrounds a central opening **232**. The plate member **231** comprises a first side edge **233** and a second side edge **234**. The plate member **231** may be elongated in a direction between the first and second side edges **233**, **234**. The plate member **231** may have a planar top surface **235** and a planar bottom surface **245**.

(73) The plate member **231** may comprise a first linear section **236** which comprises the first side edge **233** and a second linear section **237** which comprises the second side edge **234**. The plate member **231** may further comprise a third linear section **238** and a fourth linear section **239** each extending between the first and second linear sections **236**, **237**. The plate member **231** may comprise a plurality of apertures **240a-d** including a first aperture **240a** located along the first linear section **236** adjacent to the third linear section **238**, a second aperture **240b** located along the first linear section **236** adjacent to the fourth linear section **239**, a third aperture **240c** located along the second linear section **237** adjacent to the third linear section **238**, and a fourth aperture **240d** located along the second linear section **237** adjacent to the fourth linear section **239**. Each of the apertures **240a-d** may be aligned with the one of the apertures **223** in one of the first and second plugs **210**, **220** when the adaptor assembly **200** is assembled onto the housing **100** so that a fastener (e.g., a screw) can be inserted through each aligned pair of one of the apertures **240a-d** and one of the apertures **223**.

(74) The adaptor assembly **200**, and more specifically the bracket **230** thereof, may comprise a second connection feature that is configured for coupling the sensor apparatus **1000** to a second type of animal collar as described further below. The bracket **230** may comprise a first connector portion **241** extending obliquely from the first side edge **233** of the plate member **231** and a second connector portion **242** extending obliquely from the second side edge **234** of the plate member **231**. Each of the first and second connector portions **241**, **242** may extend in a direction away from the bottom surface **245** of the plate member **231** so that an inner surface of the connector portions **241**, **242** forms an oblique angle which is greater than 90° with the bottom surface **245** of the plate member **231**. In alternative embodiments, the first and second connector portions **241**, **242** may extend perpendicularly from the plate member **231**, although the oblique embodiment may have a more conformal fit with the contours of the animal's neck.

(75) The first connector portion **241** may comprise a first elongated aperture **243**. The second connector portion **242** may comprise a second elongated aperture **244**. The elongated apertures **243**, **244** may be elongated in a direction between a front edge **246** and a rear edge **247** of the bracket **230**. The first and second connector portions **241**, **242**, and more specifically the first and second elongated apertures **243**, **244** thereof, may form the second connection feature of the adaptor assembly **200**.

(76) Referring to FIGS. **8A-8C** in succession, the attachment or coupling of the adaptor assembly **200** to the housing **100** will be described. It should be appreciated that in some embodiments the sensor apparatus **1000** may be a kit which comprises the housing **100** and the adaptor assembly **200**. It may be up to the user or consumer to decide whether to attach the adaptor assembly **200** to the housing **100** or not, depending on whether the first or second type of collar is intended to be used with the sensor apparatus **1000**. The sensor apparatus **1000** may comprise the housing **100** with the electronic circuit **300** contained therein, or the sensor apparatus **1000** may comprise the housing **100** with the electronic circuit **300** contained therein and with the adaptor assembly **200** coupled thereto. These are both different forms of the sensor apparatus **1000** to be used in accordance with the type of collar to which the sensor apparatus **1000** is to be attached.

(77) The adaptor assembly **200** may be sold together with the housing **100** and electronic circuit **300**. Alternatively, the housing **100** and the electronic circuit **300** may be sold together, and the adaptor assembly **200** may be an auxiliary part that is sold separately from the housing **100** and the electronic circuit **300** when desired to use the sensor apparatus **1000** with a second type of animal collar.

(78) With reference to FIG. **8A**, the first step in the coupling of the adaptor assembly **200** to the housing **100** may comprise inserting the first plug **210** into the first nesting region **175** of the housing **100** and inserting the second plug **220** into the second nesting region **195** of the housing **100**. This may comprise translating or moving the first and second plugs **210**, **220** downwardly into the first and second nesting regions **175**, **195**, respectively. In doing this, the body portion of the first plug **210** is positioned within the first nesting region **175** and the protrusion **221** of the first plug **210** is positioned within the gap **174** between the first and second arms **167**, **169**. Furthermore, the ribs **222** of the first plug **210** mate with the gripping protrusions **173** on the inner surfaces of the first and second arms **167**, **169**. FIG. **1** illustrates the second plug **220** nesting with the second nesting region **195** and also shows the position of the protrusion of the second plug **220** within the gap **194** between the arms **187**, **189**. The same structural arrangement is achieved with the first plug **210** in the first nesting region **175**. As noted herein, in some embodiments the first and second plugs **210**, **220** may not be used and instead the mounting bracket **230** may be mounted directly to the housing **100**.

(79) The first and second plugs **210**, **220** may maintain a flush exterior of the device. Furthermore, as seen in FIG. **1**, the first and second plugs **210**, **220** may completely block the first and second nesting regions **175**, **195**, thereby preventing the first type of collar **10** from utilizing this space for attachment to the sensor apparatus **1000**. Thus, in this embodiment, the first connection feature **150** (which comprises the first and second nesting regions **175**, **195**) is inaccessible for coupling the sensor apparatus **1000** to the first type of animal collar **10** when the plugs **210**, **220** are coupled to the housing **100**. When the first and second plugs **210**, **220** are coupled to the housing **100**, the downward facing shoulder **225** of the first plug **210** engages or contacts an upper edge of the first and second arms **167**, **169** (or the third and fourth arms **187**, **189** for the second plug **220**). This engagement forms a stop feature which prevents the first plug **210** from passing fully through the first nesting region **175** (and prevents the second plug **220** from passing fully through the second nesting region **195**). That is, the abutting contact between the downward facing shoulder **225** of the first plug **210** and the first and second arms **167**, **169** prevents the first plug **210** from moving/translating any further. The first plug **210** can only be removed from the first nesting region **175** by moving/translating the first plug **210** upwardly back from where it came.

(80) FIG. 8B illustrates the housing **100** with the first and second plugs **210**, **220** positioned within the first and second nesting regions **175**, **195** as previously described. When so positioned, the bottom surface **214** of the first and second plugs **210**, **220** are exposed. As such, the apertures **223** in the bottom **214** of the first and second plugs **210**, **220** are also exposed and accessible. Next, the bracket **230** is positioned adjacent to the bottom of the housing **100** and moved towards the housing **100** until the first and second linear sections **236**, **237** of the bracket **230** abut against the bottom **214** of the first and second plugs **210**, **220**, respectively. Once the bracket **230** is so positioned, the apertures **240a-d** in the bracket **230** are aligned with the apertures **223** in the plugs **210**, **220**. Next, the fasteners **201** (e.g., screws) may be inserted through the aligned apertures **240a-d**, **223** to couple the bracket **230** to the first and second plugs **210**, **220**.

(81) As discussed herein, the first and second plugs **210**, **220** may be omitted in some embodiments and the bracket **230** may be coupled directly to the housing **100**. In other embodiments, the first and second plugs **210**, **220** may be included and attached to the housing **100** as described herein, and the bracket **230** may be coupled to the housing **100** instead of to the first and second plugs **210**, **220**. For example, the housing **100** may have screw apertures or negative space formed therein for receiving the screws **201** to mount or couple the bracket **230** to the housing **100**. The location of the holes **240a-d** along the bracket **230** may be altered from that shown to ensure alignment between the holes **240a-d** and any screw apertures formed into the housing **100**. In an embodiment whereby the bracket **230** is coupled directly to the housing **100**, the first and second plugs **210**, **220** may be retained in place via friction or tight fit against the housing **100**, adhesive, mechanically engaging features, indent/detent type engagement, or the like.

(82) At this point, the adaptor assembly **200** is coupled to the housing **100** and cannot be separated from the housing **100** without first removing the fasteners **201**. This is shown in FIG. 8C. As previously mentioned, the first and second plugs **210**, **220** are prevented from moving further downwardly due to the abutting contact between the downwardly facing shoulder **225** of the first and second plugs **210**, **220** and the arm structures of the first and second connectors **160**, **180**. Furthermore, the bracket **230** prevents the attachment structure **200** from moving in the opposite direction (i.e., towards the top **105** of the housing **100**) due to its abutment against the bottom of the housing **100**.

(83) FIG. 9 illustrates a second type of animal collar **20** coupled to the sensor apparatus **1000** when the sensor apparatus **1000** includes the adaptor assembly **200** coupled to the housing **100**. The second type of animal collar **20** may differ from the first type of animal collar **10** in at least one characteristic. The at least one characteristic may be material. The second type of animal collar **20** may be formed from metal and the first type of animal collar **10** may not be formed from metal. The second type of animal collar **20** may be chew resistant, due to its being formed from metal, for example. The second type of animal collar **20** may comprise a plurality of metal linkages **21** that are attached together. Each of the linkages **21** may include a hook portion and an apertured portion, such that the hook portion of one linkage **21** may be inserted into the apertured portion of an adjacent linkage **21**.

(84) The second type of animal collar **20** may have a first end linkage **22** and a second end linkage **23**. The hook portion of the first end linkage **22** may be inserted into the first elongated aperture **243** of the first connector portion **241** of the adaptor assembly **200**. The hook portion of the second end linkage **23** may be inserted into the second elongated aperture **244** of the second connector portion **242** of the adaptor assembly **200**. The first and second end linkages **22**, **23** may be detachable from the adaptor assembly **200** by pivoting to remove the hook portions thereof from the first and second elongated apertures **243**, **244**. The second type of animal collar **20** may further include a clasp mechanism **25**, which may be a breakaway mechanism to prevent choking, a buckle-style mechanism, a clasp, or any other type of mechanism which allows a person to attach and detach the second type of animal collar **20** from around an animal's neck.

(85) FIGS. 10 and 11 illustrate an alternative embodiment of a sensor apparatus **2000** in accordance

with the invention disclosed herein. The sensor apparatus **2000** comprises a housing **400** and an adaptor assembly **410** with the adaptor assembly **410** comprising first and second plugs **420** (only one is visible, but the description of the first and second plugs **210**, **220** of the prior embodiment is applicable) and a bracket **430**. The sensor apparatus **2000** is identical to the sensor apparatus except that the bottom surface of the housing **400** is flat rather than arcuate/concave. Other than this, the description of the sensor apparatus **1000** is entirely applicable to the sensor apparatus **2000**.

(86) FIGS. **12-14** illustrate yet another embodiment of a sensor apparatus **3000** in accordance with the invention disclosed herein. The sensor apparatus **3000** comprises a housing **500** and an adaptor assembly **600**. The housing **500** is identical to the housing **100** previously described except for the addition of slots in the inner surfaces of the flanges of the connection feature thereof, described further below. Other than the slots, the housing **500** is completely identical to the housing **100** and thus the description of the housing **100** is applicable. Furthermore, the sensor apparatus **3000** (and all other sensor apparatuses described herein) includes an electronic circuit which is identical to the electronic circuit **300** although it is not depicted or described specifically with regard to this embodiment. It should be understood that each sensor apparatus described herein includes an electronic circuit and the description of the electronic circuit **300** is therefore applicable to each embodiment described herein.

(87) As with the previous embodiment, the housing **500** comprises a main body portion **501** having a first end **502** and a second end **503** and a first connection feature **510**. The first connection feature **510** comprises a first connector **511** located at the first end **502** of the main body portion **501** and a second connector (not visible) located at the second end **503** of the main body portion **501**. The second connector is not shown in the drawings, but is identical to the first connector **511** and thus the description of the first connector **511** is applicable to the second connector.

(88) The first connector **511** comprises a first flange **512** and a second flange **513** extending from the first end **502** of the main body portion **501** of the housing **500** in a direction generally perpendicular to the first end **502**. The first flange **512** may be adjacent to a front of the housing **500** and the second flange **513** may be adjacent to the rear of the housing **500**. The first flange **512** may have an inner surface **514** and the second flange **513** may have an inner surface **515**. The inner surfaces **514**, **515** may face each other. The first connector **511** may further comprise a first arm **516** extending from a terminal end of the first flange **512** in a direction towards the second flange **513** and a second arm **517** extending from a terminal end of the second flange **513** in a direction towards the first flange **512**. The structure of the first and second flanges **512**, **513** and the first and second arms **516**, **517** is identical to the structure of the same parts noted above with reference to the sensor apparatus **1000** and thus any additional description of those parts provided above is applicable to this embodiment as well.

(89) The difference between this embodiment and the one previously described is as follows. A first slot **520** is formed into the inner surface **514** of the first flange **512** and a second slot **521** is formed into the inner surface **515** of the second flange **513**. The first and second slots **520**, **521** may be elongated in a vertical direction. The first slot **520** may extend from a top edge **522** of the first flange **512** to a floor **523** that is spaced above the bottom edge **524** of the first flange **512**. The second slot **521** may extend from a top edge **525** of the second flange **513** to a floor (not visible, but the same as the floor **523**) that is spaced above the bottom edge **526** of the second flange **513**. Thus, the first and second slots **520**, **521** may be open at the top edges **522**, **525** of the first and second flanges **512**, **513**. The main body **501** of the housing **500** also has a second connector at the second end **503** which has an identical structure including the flanges, the arms, and the slots as described.

(90) The adaptor assembly **600** comprises two identical attachment structures **605**, **606**, one for attachment to the first connector **511** on the first side of the main body **501** and one for attachment to the second connector (not visible) on the second side of the main body **501**. Thus, the adaptor assembly **600** comprises a first pin **610**, a first plate member **620** configured to be coupled to the

first pin **610**, a second pin **630**, and a second plate member **640** configured to be coupled to the second pin. The first pin **610** and the first plate member **620** form the first attachment structure **605** and the second pin **630** and the second plate member **640** form the second attachment structure **606**.

(91) The first and second pins **610**, **630** are elongated cylindrical shaped structures which may be formed from metal, or alternatively plastic or other rigid materials. The first plate member **620** comprises a loop portion **621** and a plate portion **622** extending from the loop portion **621**. The first plate member **620** comprises a front surface **626**, a rear surface **627**, a first side edge **628**, and a second side edge **629**. The loop portion **621** defines an elongated through-hole **623** that extends from the first side edge **628** to the second side edge **629**. The first pin **610** is positioned within the elongated through-hole **623** when the first pin **610** is coupled to the first plate member **620**. A first end portion **611** of the first pin **610** protrudes from the first side edge **628** and a second end portion **612** of the first pin **610** protrudes from the second side edge **629** when the pin **610** is positioned within the through-hole **623**. The plate portion **622** comprises an aperture **624** that extends from the front surface **626** to the rear surface **627**. The aperture **624** may be in the shape of an elongated oval. The first plate member **620** may be an integral structure formed from sheet metal which is bent into the shape as shown in the drawings.

(92) The second plate member **640** comprises a loop portion **641** and a plate portion **642** extending from the loop portion **641**. The first plate member **640** comprises a front surface **646**, a rear surface **647**, a first side edge **648**, and a second side edge **649**. The loop portion **641** defines an elongated through-hole **643** that extends from the first side edge **648** to the second side edge **649**. The second pin **630** is positioned within the elongated through-hole **643** when the second pin **630** is coupled to the second plate member **640**. A first end portion **631** of the second pin **630** protrudes from the first side edge **648** and a second end portion **632** of the second pin **630** protrudes from the second side edge **649** when the pin **630** is positioned within the through-hole **643**. The plate portion **642** comprises an aperture **644** that extends from the front surface **646** to the rear surface **647**. The aperture **644** may be in the shape of an elongated oval. The second plate member **640** may be an integral structure formed from sheet metal which is bent into the shape as shown in the drawings.

(93) The adaptor assembly **600** is coupled to the housing **500** as follows. First, each attachment structure **605**, **606** is assembled by inserting the first and second pins **610**, **630** through the through-holes of the respective first and second plate members **620**, **640**. The first and second pins **610**, **630** have lengths that are greater than the lengths of the through-holes **623**, **643** so that first and second end portions **611**, **612**, **631**, **632** of the first and second pins **610**, **630** protrude from the first and second ends of the first and second plate members **620**, **640**. The first and second end portions **611**, **612**, **631**, **632** are then inserted into the slots previously described. In particular, with reference to the first connector **511** only because the second connector is not visible in the drawings, the first end portion **611** of the first pin **610** is inserted into the slot **521** and simultaneously the second end portion **612** of the first pin **610** is inserted into the slot **520**. The first pin **610** is then moved downwardly within the slots **520**, **521** until the first and second end portions **611**, **612** abut against the floors **523** of the slots **520**, **521**. The pins **610**, **620** may lock in place once so positioned. This may be achieved with a spring feature on the pins **610**, **630**, with the slots **520**, **521** having a recess located adjacent to the floor **523** thereof, or with the slots being extended to have an angled portion at the bottom of the slot (the angled portion may extend at an angle relative to the remainder of the slot to prevent the pin from falling out). Other techniques for achieving this locking engagement are also contemplated.

(94) When so assembled, the first and second plate members **620**, **640** are pivotable relative to the first and second pins **610**, **630** to which they are coupled. The first plate member **620** hangs from the first pin **610** and is located within the first nesting region. Thus, the first plate member **620** is positioned within the first nesting region of the housing and is capable of pivoting therein. The second plate member **640** hangs from the second pin **630** and is located within the second nesting

region. Thus, the second plate member **640** is positioned within the second nesting region of the housing and is capable of pivoting therein.

(95) Referring to FIG. **15**, the sensor apparatus **3000** may be coupled to the second type of collar **20** in a manner similar to that which was described above. In particular, the end linkages **22**, **23** may engage the apertures **624**, **644** of the first and second plate members **620**, **640** of the first and second attachment structures **605**, **606**. Of course, if it is desired to use the sensor apparatus **3000** with the first type of animal collar **10**, then the first and second attachment structures **605**, **606** of the adaptor assembly **600** is removed/separated from the housing **500** and then the first type of animal collar **10** is coupled directly to the housing **500** in the same manner as discussed above with reference to FIGS. **5A** and **5B**. Thus, the adaptor assembly **600** can be attached to and/or separated from the housing **500** depending on whether the first type of animal collar **10** or the second type of animal collar **20** is to be used.

(96) Referring to FIGS. **16-18**, another embodiment of a sensor apparatus **4000** is illustrated in accordance with an embodiment of the present invention. The sensor apparatus **4000** comprises a housing **700** and an adaptor assembly **800** that is configured to be coupled to the housing **700**. The housing **700** is identical to the housing **100** except for the differences described herein. Thus, the description of the housing **100** is applicable to the housing **700** with the minor modifications described herein which are intended to facilitate coupling of the adaptor assembly **800** to the housing **700**.

(97) In this embodiment, the housing **700** includes a main body **701** and a connection feature which includes a first connector **702** on a first side of the main body **701** and a second connector **703** on a second side of the main body **701**. The first connector **702** comprises first and second flanges **710**, **712** and first and second arms **711**, **713** on the first side of the main body **701**. The second connector **703** comprises third and fourth flanges **730**, **732** and third and fourth arms **731**, **733** on the second side of the main body **701**. The modification of the housing **700** as compared to the housing **100** is the addition of a first hole **750** extending through the first flange **710**, a second hole **751** extending through the second flange **712**, a third hole **752** extending through the third flange **730**, and a fourth hole **753** extending through the fourth flange **732**.

(98) The adaptor assembly **800** comprises a first rod **810** and a second rod **820**. Each of the first and second rods **810**, **820** are elongated cylinder-shaped structures. The first rod **810** has end portions that are located within the first and second holes **750**, **751** and the rest of the first rod **810** spans across the first nesting region. The first rod **810** may extend across the first nesting region at a location below the first and second arms **711**, **713**. The second rod **820** has end portions that are located within the third and fourth holes **752**, **753**. The rest of the second rod **820** spans across the second nesting region. The second rod **820** may extend across the second nesting region at a location below the third and fourth arms **731**, **733**.

(99) FIG. **19** illustrates the second type of collar **20** coupled to the first and second rods **810**, **820** of the adaptor assembly **800** of the sensor apparatus **4000**. In this embodiment, it may also be possible to couple the sensor apparatus **4000** to the first type of collar **10** using the connection features of the housing **700** even while the adaptor assembly **800** is coupled to the housing **700**. That is, the adaptor assembly **800** may not impede the ability of attaching the first type of collar **10** to the housing **700** in the manner described above with reference to FIGS. **5A** and **5B**.

(100) Referring to FIGS. **20** and **21**, a sensor apparatus **5000** is illustrated. The sensor apparatus **5000** comprises a housing **900** containing an electrical circuit as described previously. The housing **900** is identical to the housing **100**, except that there is a charging port **901** located on the exterior of the housing **900**. The charging port **901** may be a USB port or any other type of port intended to accept a charging adaptor plug. The charging port **901** may be located within a recess **902** formed in the housing **900**.

(101) The sensor apparatus **5000** further comprises a band **910** that is configured to fit around the housing **900** to cover the charging port **901**. The band may be formed from an elastomeric material,

such as silicone. The band **910** may have a color that is outside of the visible spectrum of dogs. For example without limitation, the band **910** may be magenta, red, or cyan. The band **910** may be a loop-shaped structure that can wrap fully around a portion of the housing **900** when coupled thereto. Thus, the band **910** may be configured to cover the charging port **901** when the band **910** is coupled to the housing **900**. The band **910** may include a nub designed to fit into the charging port **901** to hold the band in place **910** and keep debris out of the charging port **901**. The band **910** may be elastomeric so as to be stretchy enough to fit around the housing **900** but stable enough to be able to slide along the housing **900** to expose the charging port **901** when charging is desired.

(102) FIGS. **22** and **23** illustrate the sensor apparatus **5000** with a minor modification to the housing **900**. In particular, the housing **900** comprises an annular recess **920** where the band **910** is intended to be placed when the band **910** is coupled to the housing **900**. Thus, when the band **900** is coupled to the housing **900** with this modification, the outer surface of the band **910** is flush with the outer surface of the housing **900**. This results in a seamless aesthetic for the finished product. The band **910** as shown in FIGS. **21-24** may be incorporated into any embodiment of the sensor apparatus as described herein.

(103) While the inventions have been described with respect to specific examples including presently preferred modes of carrying out the inventions, those skilled in the art will appreciate that there are numerous variations and permutations of the above described systems and techniques. It is to be understood that other embodiments may be utilized and structural and functional modifications may be made without departing from the scope of the present inventions. Thus, the spirit and scope of the inventions should be construed broadly as set forth in the appended claims.

Claims

1. A sensor apparatus configured for attachment to a collar worn by an animal, the sensor apparatus comprising: a sensor assembly comprising: a housing comprising a first connection feature configured for coupling the sensor assembly to a first type of animal collar, the housing defining a cavity; and an electronic circuit located within the cavity of the housing, wherein the electronic circuit is configured to at least one of store information and collect data; and an adaptor assembly configured to be coupled to the sensor assembly, the adaptor assembly comprising a second connection feature configured for coupling the sensor apparatus to a second type of animal collar that is different from the first type of animal collar.
2. The sensor apparatus according to claim 1 wherein the first connection feature is integral with the housing, and wherein when the adaptor assembly is coupled to the housing, the first connection feature is inaccessible for coupling the sensor apparatus to the first type of animal collar.
3. The sensor apparatus according to claim 1 wherein the electronic circuit comprises a power source and at least one sensor for collecting data related to the animal.
4. The sensor apparatus according to claim 1 wherein the second type of animal collar is formed from metal and the first type of animal collar is not formed from metal, wherein the first type of animal collar is formed from a material selected from the group consisting of nylon, polyester, neoprene, rubber, elastomer, and leather.
5. The sensor apparatus according to claim 1 wherein the housing comprises a charging port for charging a power source of the electronic circuit, and further comprising a band coupled to the housing to cover the charging port, wherein the band is movable relative to the housing to expose the charging port for charging of the power source.
6. The sensor apparatus according to claim 5 wherein the band is formed from an elastomeric material and wherein the band wraps around a portion of the housing.
7. The sensor apparatus according to claim 1 wherein the housing comprises a main body portion having a first end and a second end, wherein the first connection feature comprises a first arm structure that is spaced apart from the first end of the main body portion of the housing by a first

nesting region and a second arm structure that is spaced apart from the second end of the main body portion of the housing by a second nesting region.

8. The sensor apparatus according to claim 7 wherein the adaptor assembly comprises: a first plug configured for insertion into the first nesting region; a second plug configured for insertion into the second nesting region; and a bracket configured to be coupled to either: (1) each of the first and second plugs; or (2) the housing; and wherein the bracket comprises the second connection feature.

9. The sensor apparatus according to claim 7 wherein the adaptor assembly comprises: a first rod extending across the first nesting region at a position between the first arm structure and a bottom end of the housing; and a second rod extending across the second nesting region at a position between the second arm structure and the bottom end of the housing; and wherein the second type of animal collar is configured to engage the first and second rods to couple the sensor apparatus to the second type of animal collar.

10. The sensor apparatus according to claim 7 further comprising: wherein the first connection feature comprises: a first flange and a second flange extending from the first end of the main body portion of the housing and having inner surfaces that face one another, the first arm structure extending between the first and second flanges; a third flange and a fourth flange extending from the second end of the main body portion of the housing and having inner surfaces that face one another, the second arm structure extending between the third and fourth flanges; and wherein the inner surfaces of each of the first, second, third, and fourth flanges comprises a slot; and wherein the adaptor assembly comprises: a first pin disposed within the slots of the first and second flanges and extending across the first nesting region; and a first plate member coupled to the first pin and at least partially located within the first nesting region, the first plate member comprising a first aperture; a second pin disposed within the slots of the third and fourth flanges and extending across the second nesting region; and a second plate member coupled to the second pin and at least partially located within the second nesting region, the second plate member comprising a second aperture; and wherein the second type of animal collar is configured to mate with the first and second apertures to couple the sensor apparatus to the second type of animal collar.

11. The sensor apparatus according to claim 10 wherein the first plate member is pivotable relative to the first pin within the first nesting region and the second plate member is pivotable relative to the second pin within the second nesting region.

12. The sensor apparatus according to claim 1 further comprising: the housing comprising a main body portion having a first end and a second end; the first connection feature comprising: a first connector extending from the first end of the main body portion so that a first nesting region is defined between the first connector of the first connection feature and the first end of the housing; and a second connector extending from the second end of the main body portion so that a second nesting region is defined between the second connector of the first connection feature and the second end of the housing; and wherein when the sensor apparatus is coupled to the first type of animal collar, a first portion of the first type of animal collar nests within the first nesting region, a second portion of the first type of animal collar nests within the second nesting region, and a third portion of the first type of animal collar located between the first and second portions of the first type of animal collar extends along a bottom of the housing.

13. The sensor apparatus according to claim 12 wherein the adaptor assembly comprises: a first plug configured for insertion into the first nesting region; a second plug configured for insertion into the second nesting region; and a bracket configured to be coupled to either: (1) each of the first and second plugs; or (2) the housing; and wherein the bracket comprises the second connection feature.

14. The sensor apparatus according to claim 13 wherein the bracket comprises a plate member that is adjacent to a bottom of the housing when the adaptor assembly is coupled to the housing, a first connector portion extending obliquely from a first side edge of the plate member and comprising a first aperture, and a second connector portion extending obliquely from a second side edge of the

plate member and comprising a second aperture, wherein the second type of animal collar is configured to mate with the first and second apertures to couple the sensor apparatus to the second type of animal collar.

15. The sensor apparatus according to claim 13 further comprising: wherein the first connector comprises: a first flange extending from the first end of the main body portion adjacent to a front of the main body portion; a second flange extending from the first end of the main body portion adjacent to a rear of the main body portion; a first arm extending from the first flange towards the second flange, the first arm terminating in a first distal end; and a second arm extending from the second flange towards the first flange, the second arm terminating in a second distal end that is spaced apart from the first distal end by a first gap; and wherein the second connector comprises: a third flange extending from the second end of the main body portion adjacent to the front of the main body portion; a fourth flange extending from the second end of the main body portion adjacent to the rear of the main body portion; a third arm extending from the third flange towards the fourth flange, the third arm terminating in a third distal end; and a fourth arm extending from the fourth flange towards the third flange, the fourth arm terminating in a fourth distal end that is spaced apart from the third distal end by a second gap.

16. The sensor apparatus according to claim 15 wherein the first plug comprises a first body portion that nests within the first nesting region and a first protrusion that nests within the first gap and wherein the second plug comprises a second body portion that nests within the second nesting region and a second protrusion that nests within the second gap.

17. A wearable animal information apparatus comprising: a sensor apparatus comprising: a housing comprising a first connection feature; an adaptor assembly configured to be coupled to the housing, the adaptor assembly comprising a second connection feature; and an electronic circuit located within a cavity of the housing, wherein the electronic circuit is configured to at least one of store information and collect data; a first type of collar configured to be coupled to the sensor apparatus utilizing the first connection feature of the housing, wherein the first type of collar is incapable of attachment to the second connection feature of the adaptor assembly; and a second type of collar configured to be coupled to the sensor apparatus utilizing the second connection feature of the adaptor assembly, wherein the second type of collar is incapable of attachment to the first connection feature of the housing.

18. The wearable animal information apparatus according to claim 17 wherein the first type of collar is prevented from being coupled to the sensor apparatus utilizing the first connection feature when the adaptor assembly is coupled to the housing.

19. The wearable animal information apparatus according to claim 17 wherein the first type of collar is formed from a material selected from the group consisting of nylon, polyester, neoprene, rubber, elastomer, and leather and wherein the second type of collar is formed from metal, wherein the first connection feature comprises a pair of nesting regions within which portions of the first type of collar are positioned when the first type of collar is coupled to the sensor apparatus utilizing the first connection feature of the housing, and wherein at least a portion of the adaptor assembly is located within the pair of nesting regions when the adaptor assembly is coupled to the housing.

20. The wearable animal information apparatus according to claim 17 wherein the first connection feature comprises a pair of nesting regions within which portions of the first type of collar are positioned when the first type of collar is coupled to the sensor apparatus utilizing the first connection feature of the housing, and wherein the adaptor assembly further comprises: a first plug located within the first nesting region when the adaptor assembly is coupled to the housing; a second plug located within the second nesting region when the adaptor assembly is coupled to the housing; and a bracket located along a bottom of the housing and configured to be coupled to either: (1) each of the first and second plugs; or (2) the housing; and wherein the bracket comprises the second connection feature.
